

IHTC Metamodel

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METAMODEL SUMMARY

Overview

This metamodel captures the current IHTC work in MDEO for two connected optimisation layers: patient scheduling and nurse assignment. The scheduling side decides which patients are admitted, on what day, in which room, and in which operating theatre. The nurse-assignment side then represents room-shift demand and assigns nurses to those shifts. The model also includes day-based availability objects so that transformations can update and validate state over time.

Enumerations

- **Gender.**
Purpose: stores patient sex as M or F.
 - **AgeGroup.**
Purpose: stores patient age category as BABY, CHILD, YOUNG, ADULT, ELDERLY, or EMPTY.
Nuance: EMPTY is also used in RoomAvailability as a sentinel value for a room-day that currently has no occupants.
 - **OptimisationPhase.**
Purpose: separates the one-way search into PATIENTS and NURSES phases.
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Root Container

- **HospitalInstance.**
Purpose: root object that collects the full optimisation state for one hospital instance.
 - **Attribute:** decisionHorizon : int
 - **Link:** patients[0..*] -> Patient
 - **Link:** operatingtheatres[0..*] -> OperatingTheatre
 - **Link:** surgeons[0..*] -> Surgeon
 - **Link:** rooms[0..*] -> Room
 - **Link:** nurses[0..*] -> Nurse
 - **Link:** surgeonAvailabilities[0..*] -> SurgeonAvailability
 - **Link:** operatingTheatreAvailabilities[0..*] -> OperatingTheatreAvailability
 - **Link:** roomAvailabilities[0..*] -> RoomAvailability
 - **Link:** hospitalisationShifts[0..*] -> HospitalisationShift
 - **Link:** nurseWorkingShifts[0..*] -> NurseWorkingShift
 - **Link:** roomShiftAssignments[0..*] -> RoomShiftAssignment

- **Link:** deletedAdmissionsTrackers[0..*] -> DeletedAdmissionsTracker
- **Link:** optimisationState[1..1] -> OptimisationState
- **Link:** admissions[0..*] -> Admission

Nuance: the decisionHorizon attribute defines the number of days in scope for the planning problem.

Patient Scheduling Core

- **Patient.**

Purpose: represents a patient that may or may not be admitted.

- **Attribute:** id : int
- **Attribute:** isMandatory : boolean
- **Attribute:** isScheduled : boolean
- **Attribute:** dueDate : int
- **Attribute:** releaseDate : int
- **Attribute:** ageGroup : AgeGroup
- **Attribute:** surgeryDuration : int
- **Attribute:** gender : Gender
- **Attribute:** stayLength : int
- **Link:** assignedSurgeonId[1..1] -> Surgeon
- **Link:** incompatibleRooms[0..*] -> Room
- **Link:** dayDemand[1..*] -> PatientDayDemand

Nuance: isScheduled is currently used as a practical modelling flag to simplify matching and transformation logic.

- **Admission.**

Purpose: records the actual scheduling decision for a patient.

- **Attribute:** admissionDay : int
- **Link:** patientId[1..1] -> Patient
- **Link:** roomId[1..1] -> Room
- **Link:** operationTheatreId[1..1] -> OperatingTheatre

Nuance: the admission stores only the start day, so full-stay effects are handled through day-based support objects such as RoomAvailability.

- **Room.**

Purpose: represents a recovery room used for post-operative stays.

- **Attribute:** id : int
- **Attribute:** maxCapacity : int

- **OperatingTheatre.**

Purpose: represents a theatre resource for surgery.

- **Attribute:** id : int

- **Surgeon.**

Purpose: represents a surgeon resource.

- **Attribute:** id : int

Nuance: surgeons are assigned to patients before optimisation rather than chosen by the transformations.

Day-Based Resource State

- **SurgeonAvailability.**

Purpose: stores how much operating time a surgeon has on a given day.

- **Attribute:** day : int
- **Attribute:** maxOperatingTime : int
- **Link:** surgeonId[1..1] -> Surgeon

- **OperatingTheatreAvailability.**

Purpose: stores available theatre capacity for one theatre on one day.

- **Attribute:** day : int
- **Attribute:** maxCapacity : int
- **Link:** operatingTheatreId[1..1] -> OperatingTheatre

- **RoomAvailability.**

Purpose: stores room occupancy state for one room on one day.

- **Attribute:** day : int
- **Attribute:** occupiedBeds : int
- **Attribute:** ageGroup : AgeGroup
- **Attribute:** roomNumber : int
- **Link:** roomId[1..1] -> Room

Nuance: this class is central to enforcing room capacity and no-age-group-mixing constraints across a patient's full stay. The room number is duplicated here as a convenience field for transformation logic.

Nurse Assignment Layer

- **HospitalisationShift.**

Purpose: represents a structural room-day-shift slot to which a nurse may be assigned.

- **Attribute:** day : int
- **Attribute:** shift : int
- **Link:** room[1..1] -> Room

Nuance: workload and required skill are derived from current **Admission** and **PatientDayDemand** records rather than stored on this class.

- **Nurse.**

Purpose: represents a nurse resource.

- **Attribute:** id : int
- **Attribute:** skillLevel : int

- **NurseWorkingShift.**

Purpose: stores when a nurse is available to work and how much workload they can cover.

- **Attribute:** day : int
- **Attribute:** shift : int
- **Attribute:** maxLoad : int
- **Link:** nurse[1..1] -> Nurse

- **RoomShiftAssignment.**

Purpose: records the decision to assign a nurse to a particular hospitalisation shift.

- **Link:** nurse[1..1] -> Nurse

– **Link:** hospitalisationShift[1..1] -> HospitalisationShift

Nuance: this is the final decision object for the nurse-assignment optimisation stage.

Demand and Tracking Support

- **PatientDayDemand.**

Purpose: describes the nursing demand a patient contributes at a relative day and shift of their stay.

- **Attribute:** relativeDay : int
- **Attribute:** shift : int
- **Attribute:** workloadProduced : int
- **Attribute:** skillLevelRequired : int
- **Link:** patient[1..1] -> Patient

Nuance: together with an admission, this maps patient demand to a room-day-shift slot.

- **DeletedAdmissionsTracker.**

Purpose: counts how many admission removals have been performed during search.

- **Attribute:** count : int

Nuance: it records deletion operations during search; the active removed-patient objective separately counts mandatory patients without admissions.

- **OptimisationState.**

Purpose: stores the current **OptimisationPhase** for the one-way patient-then-nurse search.

- **Attribute:** phase : OptimisationPhase

Nuance: patient transformations apply in **PATIENTS**; nurse transformations apply in **NURSES** after mandatory patients are scheduled.

- **Occupant.**

Purpose: represents a patient-room occupancy relation with a stay length.

- **Attribute:** stayLength : int
- **Link:** patientId[1..1] -> Patient
- **Link:** assignedRoomId[1..1] -> Room

Nuance: in the current model, most practical occupancy reasoning is handled through **Admission** and **RoomAvailability**, so this class is more auxiliary than central.

Modelling Notes

- **Decision versus derived state.** An **Admission** is the patient-scheduling decision. Room occupancy is not inferred only when needed: pre-created **RoomAvailability** records store the mutable state for each room and day of the planning horizon.
- **Full-stay occupancy.** A patient occupies their assigned room from **admissionDay** through **admissionDay + stayLength - 1**. Patient transformations validate and update every corresponding room-day record, enforcing capacity and the no-age-group-mixing rule.
- **Two optimisation phases.** **OptimisationState** begins in **PATIENTS**. Once every mandatory patient is admitted, the state changes one way to **NURSES**, enabling nurse-assignment moves. T4–T13 are then unavailable, while T1–T3 remain available to admit further optional patients.
- **Derived nursing demand.** **HospitalisationShift** is a pre-created structural room-day-shift slot. Its workload and required skill are derived from the current **Admission** objects and matching **PatientDayDemand** records, so moving an admission automatically changes the nursing demand considered by objectives and nurse transformations.
- **MDEO support fields.** **isScheduled**, **roomNumber**, and **AgeGroup.EMPTY** support matching, full-stay updates, and age-group compatibility. **Occupant** is auxiliary; active occupancy uses **Admission** and **RoomAvailability**.